INCH-POUND

MIL-DTL-32554
1 August 2016

DETAIL SPECIFICATION

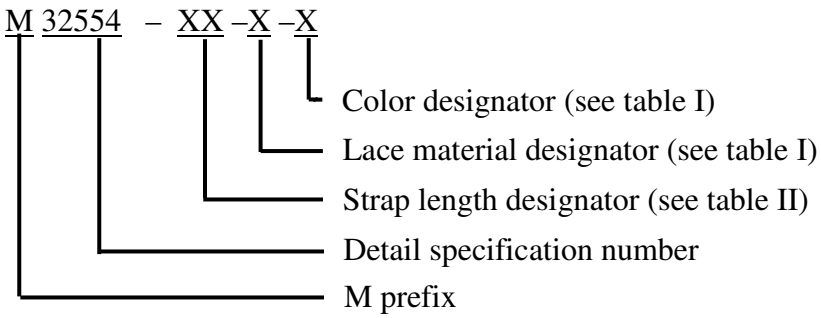
STRAPS, TIE-DOWN, ADJUSTABLE, NON-METALLIC, CABLE BUNDLING, GENERAL SPECIFICATION FOR

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design and test requirements for adjustable, non-metallic, cable bundling tie-down straps primarily intended for Electrical Wiring Interconnect System (EWIS) secondary support within aircraft applications.

1.2 Part or Identifying Number (PIN). The PINs to be used for the tie-down straps acquired in accordance to this specification are created as follows:



2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to the Naval Air Warfare Center Aircraft Division Lakehurst, Code 4.1.2, Mail Stop 120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to michael.sikora@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATION

TT-I-735	-	Isopropyl Alcohol

COMMERCIAL ITEM DESCRIPTION

A-A-52084	-	Tape, Lacing and Tying, Aramid

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-5606	-	Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance
MIL-PRF-7808	-	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-DTL-83133	-	Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34), NATO F-35, and JP-8+100 (NATO F-37)
MIL-DTL-32555	-	Tool, Installation, for Adjustable Tie-Down Straps

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202	-	Electronic and Electrical Component Parts
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) INTERNATIONAL

ASTM D5025	-	Standard Specification for Laboratory Burner Used for Small-Scale Burning Tests on Plastic Materials

(Copies of this document are available from http://www.astm.org.)

NATIONAL CONFERENCE OF STANDARDS LABORATORIES (NCSL) INTERNATIONAL

NCSL Z540.3	-	Requirements for the Calibration of Measuring and Test Equipment (DoD adopted)

(Copies of this document are available from http://www.ncsli.org.)

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SAE INTERNATIONAL

SAE AIR1351	-	Manufacturers' Identification of Aerospace Electrical and Electronic Wiring Devices and Accessories (DoD adopted)
SAE AMS2629	-	Fluid, Jet Reference (DoD adopted)
SAE AS22759/16	-	Wire, Electric, Fluoropolymer-Insulated, Extruded ETFE Medium Weight, Tin-Coated Copper Conductor, 600-Volt, 150°C, ROHS (DoD adopted)
SAE AS23190/4	-	Mounting Hardware, Cushion Clamp, Metal for Cable Harness Tying And Support, Type V, Class 1

(Copies of these documents are available online at http://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The tie-down straps furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified product list before contract award (see 4.3 and 6.3).

3.2 Tie-down straps. The tie-down straps shall consist of a length of cable lacing tape, permanently and mechanically attached to a fastener.

3.3 Materials. The materials used in the manufacture of the tie-down straps shall be such that the tie-down straps are capable of meeting all of the performance and test requirements of this specification. The lace shall be size 2, finish C, aramid aerospace cable bundling lace as specified by A-A-52084. No metal may be used in the construction of the tie-down strap assembly.

3.4 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.5 Design and construction. The design and construction of the tie-down straps shall be as specified herein. Design shall conform to all performance requirements in 3.7. The design allows for installation of tie-down straps using single, double or triple wrapping of lace around cable bundle before securing as shown on figure 1.

3.5.1 Fastener design. Fastener shall be designed to accommodate wrapping of tie-down straps around cable bundle, threading of lace through fastener lace path, tensioning of tie-down straps around cable bundle, and securing tie-down straps to cable bundle. The tie-down straps shall be secured on cable bundle with automatic activation when the installer applies 17 to 24 pounds-force (lbf) tensile force to the free end of the lace (see figure 2).

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3.6 Dimensions and tolerances. Dimensions for the various sizes of tie-down straps are as shown in table II.

3.7 Performance. The tie-down strap assembly shall conform to the requirements of this specification, when subjected to the applicable tests specified in table III.

3.7.1 Tensile strength (see 4.6.1). The tie-down strap assembly shall meet the minimum tensile strength requirements of table IV without breaking or releasing the tie-down strap from the tensile test fixture after being subjected to the tensile strength test of 4.6.1.

3.7.2 Temperature cycling. The tie-down strap assembly shall not deform, crack, break or release from the cable bundle, when subjected to the temperature cycling test of 4.6.2. Following the tensile strength test (see 4.6.1) and the temperature cycling test (see 4.6.2), the tie-down straps shall meet the minimum tensile strength after temperature cycling requirements of table V.

3.7.3 Vibration. The tie-down strap assembly shall be exposed to the vibration test of 4.6.4 without cracking, breaking, or releasing from the wire bundle.

3.7.4 Fluid immersion (see 4.6.5). The tie-down strap assembly shall be immersed in the designated fluids of table VI without cracking, distorting, breaking, separation of fastener head from lace, or releasing the locking device. The tie-down straps shall meet the tensile strength requirements of table VII and identification of product and legibility requirements of 3.7.8 upon completion of the fluid immersion test.

WARNING

The fluids listed in table VI are highly toxic to skin, eyes and respiratory tract. They are also flammable. Eye and skin protection is required. Good general ventilation is normally adequate.

3.7.5 Fuel immersion life. The tie-down strap assembly shall be immersed in designated fluids specified in 4.6.6 without cracking, distorting, breaking, separation of fastener head from lace, or releasing the locking device. The tie-down strap assembly shall meet tensile strength requirements of table VII upon completion of the fuel immersion test.

3.7.6 Slip resistance. When subjected to the slip resistance test of 4.6.7, the tie-down strap assembly shall meet the minimum slip resistance load requirements of table VIII without breaking or releasing the tie-down strap from the mandrel.

3.7.7 Flammability. The tie-down strap assembly shall be tested in accordance with the flammability test of 4.6.8. The flame shall be self-extinguishing within a maximum of 3 seconds after removal of the external flame source from the activated tie-down strap.

3.7.8 Identification of product (see 4.6.9). The manufacturing company name or symbol representing product manufacturer (see SAE AIR1351) shall be marked on the tie-down strap assembly. Identification shall be examined in accordance with 4.6.9.

3.7.9 Color. The fastener shall be black in color and the strap shall be as defined in table I.

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3.7.10 Workmanship. Workmanship shall ensure that the fastener and straps are of uniform quality, free from defects that would affect their service, and shall be uniform in appearance without any mold flash.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified herein, all inspections shall be made at ambient temperature and humidity as specified in MIL-STD-202, test conditions.

4.2.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy and quantity to permit performance of the required inspection shall be established and maintained by the contractor.

4.2.2 Tie-down strap assembly. The tie-down strap shall be installed on the cables or mandrels by the test activity using the MIL-DTL-32555 tie down strap tool.

4.3 Qualification inspection. Qualification inspection shall consist of examinations and tests specified in table III. The qualifying activity shall perform or witness the particular tests as specified in table III.

4.3.1 Qualification inspection process. For guidance on the qualification inspection process, see 6.3 and 6.3.1.

4.3.2 Qualification inspection samples. The sample lot size needed for each test for each wrap configuration (single, double, triple) of tie-down strap shall be as shown in table III.

a. For the qualifying activity tests, 55 tie-down straps (44 for testing and 11 spares) for each part number shall be separately packaged and forwarded to the address designated in the letter of authorization. Sufficient wire conforming to SAE AS22759/16 shall be provided to perform the designated tests. Each sample shall be identified by marking each package with the following information:

Sample for qualification:

Specification MIL-DTL-32554 Straps, Tie-Down, Adjustable, Non-Metallic, Cable Bundling
Part number: M32554-XX-X-X
Name of supplier
Supplier's part number and Commercial and Government Entity (CAGE) Code
Submitted (date) under authorization (reference letter authorizing the inspection)

4.3.3 Retention of qualification. Retention of qualification inspections shall be performed every 36 months after the original qualification approval date. The qualifying activity shall perform or witness the tests specified in table III. Unless otherwise specified in table III, sequential testing is required. The qualifying activity may shift the reporting date to accommodate testing schedules, but no later than 18 months from the retention of qualification due date.

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4.3.3.1 No production certification. If no production of the qualified products has occurred for the entire reporting period, the supplier shall provide certification to the qualifying activity that no changes in the product materials, design, manufacturing processes, or site of production have occurred since qualification approval. Certification for more than one reporting period is not permitted. The supplier shall contact the qualifying activity to determine the conditions for certification. The supplier shall submit the certification on DD Form 1718 provided by the qualifying activity.

4.4 Conformance inspection.

4.4.1 Inspection of product for delivery. Inspection of product for delivery shall consist of conformance inspection tests specified in table IX.

4.4.1.1 Inspection lot. An inspection lot shall consist of assemblies (fastener and lacing strap) for a particular size produced under essentially the same manufacturing conditions and presented for inspection at one time.

4.4.1.2 Sampling plan. Statistical sampling shall be in accordance with table X. A randomly selected group(s) of samples is required. For acceptance of the lot there shall be zero occurrences of defects.

4.5 Examinations.

4.5.1 Visual and dimensional examination. Tie-down straps shall be examined to verify that the materials, design, construction, and physical dimensions are in accordance with this specification. Dimensions shall be measured with calibrated devices using standard laboratory methodology.

4.5.2 Examination of product (qualification inspection). Tie-down straps shall be carefully examined to determine compliance with the requirements of this specification with respect to workmanship, product identification, and dimensions as specified in 3.6, 3.7.8, 3.7.9 and 3.7.10.

4.5.3 Examination of product (conformance inspection). For conformance inspection, examination of product shall consist of the inspections specified in table IX.

4.6 Test methods.

4.6.1 Tensile strength (see 3.7.1). The test sample shall be securely attached and positioned on the tensile test fixture as shown on figure 3. The fixture shall be positioned in a tensile testing machine and sufficient force applied to break the strap or, when applicable, to release the locking device or separate the strap from the locking device. The jaws of the tensile test machine shall move apart at a speed of one inch per minute.

4.6.2 Temperature cycling (see 3.7.2). The tie-down strap shall be mounted to the wire bundle in the position indicated by figure 4 and be subjected to the Thermal Shock Method 107, Test Condition C of MIL-STD-202. The wire bundles for tie-down straps shall contain a minimum of three SAE AS22759/16 wires American Wire Gauge (AWG) size 12 through 24. The bundle diameter shall be sufficient size to permit the straps to slide over the required tensile strength fixture mandrels. The time duration for steps 2 and 4 shall not be more than two minutes. The fastener and strap shall be inspected for cracks, distortions, breaks or releasing of the locking device, and then subjected to the tensile strength test after temperature cycling test specified in 4.6.3.

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4.6.3 Tensile strength after temperature cycling. The temperature cycling samples shall be prepared for the tensile strength test by removing the wire bundle from the test fixture then removing the center wire(s) from the bundle permitting the wire bundle to collapse. Slip the tie-down strap assembly off the collapsed bundle and remount to the test fixtures as shown on figure 3. Perform the tensile strength test specified in 4.6.1.

4.6.4 Vibration test (see 3.7.3).

4.6.4.1 Test conditions. The environmental test facility shall meet the general climate requirements described in MIL-STD-810, Section 5 General Laboratory Test Method Guidelines, Test Conditions. The test facility shall be properly equipped with regard to space and utilities for performance of the testing described in this procedure.

4.6.4.2 Test equipment. Calibration of all test measuring and data recording equipment used by the test facility shall be administered using a calibration system in compliance with NCSL Z540.3. The vibration shaker system used shall be of sufficient force and displacement capabilities to perform the vibration and shock specified in table XI.

4.6.4.3 Method of mounting. The tie-down strap configuration of single, double or triple wrap to be tested shall be installed on the wire bundle diameter specified in table XI with spacing and dimensions shown on figure 4. For each tie-down strap configuration, a minimum of 5 wire bundles with 4 tie-down straps installed shall be tested. Wire bundle assemblies shall be comprised of a minimum of three SAE AS22759/16 wires (12-24 AWG). The wire bundle may be taped on each end to prevent wire slippage during vibration. The wire bundle assemblies shall be installed flat on the vibration fixture using SAE AS23190/4 clamp sizes specified in table XI with 10-32 UNF-3A bolts of 160 ksi or higher yield strength. Bolts shall be torqued to 30 ±3 inch-pounds (in-lb). Multiple wire bundle assemblies may be operated on a test fixture simultaneously provided the capacity of the test equipment is not exceeded and the wire bundle assemblies do not dynamically interact. If multiple assemblies are tested simultaneously, a minimum of one wire bundle of each size shall have a response accelerometer affixed to each of the two SAE AS23190/4 mounting clamps as shown on figures 5 and 6. A minimum of one control accelerometer shall be affixed to the vibration fixture as shown on figures 5 and 6.

4.6.4.4 Instrumentation. The instrumentation required for this test includes an electrodynamic shaker, vibration controller, and general purpose accelerometers. Specific instrumentation shall be of sufficient capability to perform the tests herein.

4.6.4.5 Frequency response survey. The frequency response of the test article shall be measured to determine the response characteristics. This measurement is collected as part of the fixture survey to alert the test operator to possible dangerous resonance conditions that may exist with the test article. This test shall be performed with the test article installed on the wire bundle assembly and affixed to the mounting fixture. A frequency response survey for both the mounting fixture and the test article shall be completed in both the vertical and lateral axis as defined by figures 5 and 6. Fixture symmetry and other factors may be considered when conducting the fixture survey. Variations from the procedure listed below are acceptable provided that sufficient data exists to show that the frequency response of the fixture at the mounting locations is within 3 decibels (dB) of the control signal, and that the cross axis response does not exceed that of the control signal.

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4.6.4.5.1 Procedure.

a. Mount the test assemblies as specified in 4.6.4.3 on the fixture to vibration machine.

b. Attach control accelerometers adjacent to the test article on the mounting fixture as shown on figures 5 and 6.

c. Attach a triaxial accelerometer close to each wire clamp as shown on figures 5 and 6.

d. Perform a sine sweep (0.024 inch double amplitude, 1 g maximum, 5-2000 Hz, 1 octave/minute) and record the response of all accelerometers.

e. Optimize the control strategy and repeat the fixture survey if necessary.

4.6.4.6 Test method. Random vibration testing shall be conducted per in both the vertical and lateral axis as defined by figures 5 and 6. Simultaneous testing of multiple bundle assemblies is permissible. Test article assemblies for each configuration of single, double or triple wrap, shall be tested once along a single axis. Wire bundles and mounting clamps may be used for more than one test of new test article components providing the bundles are in an acceptable condition to indicate abrasion from subsequent tests or to otherwise prevent any additional variation. Following each test, the test articles shall be inspected for cracks, distortions, breaks or releasing of the locking fastener.

4.6.4.7 Failure criteria (see 3.7.3). Tie-down straps shall not fail within the duration of the test administered. Visual damage to the strap is considered a failure. This includes cracks, breaks or releasing of the lacing strap or locking fastener.

4.6.5 Fluid immersion (see 3.7.4). Individual components shall be installed on the tensile test fixture as specified on figure 3. Two samples of each type and dash number shall be immersed for four hours in each of the fluids in table VI. Upon removal from the fluids, the components shall be placed on an absorbent material and shall remain for one hour in free air at room temperature. The components shall then pass the tensile strength test of 4.6.1 and product identification legibility test of 4.6.9.

4.6.6 Fuel immersion life (see 3.7.5). Each tie-down strap assembly shall be immersed in MIL-DTL-83133 (JP-8), F-24, or SAE AMS2629 fuel for 1080 minutes without cracking, distorting, breaking, separation of the fastener head from the lace, or releasing of the locking device. Upon removal from the test fluid, the tie-down strap assembly shall be wiped dry and then remain for 1 hour in free air at room temperature. The tie-down strap assemblies shall then be subjected to the tensile strength test specified in 4.6.1.

4.6.7 Slip resistance test (see 3.7.6). Test shall be performed in a standard tensile testing machine with the specimen being pulled through a holding fixture that prevents the tie-down strap from passing through the fixture hole. Attach tie-down strap to test mandrel. The mandrel shall be gripped by the tensile machine jaws at one end and the other end of the mandrel shall be free to move through the holding fixture. Sufficient force shall be applied to cause the tie-down strap to slip on mandrel. Test machine shall move at a rate of 1 in./min. Measure force required to make tie-down strap slip on mandrel. Discontinue test when tie-down strap travels on mandrel a distance of 0.25-inch. See figure 7 for test configuration.

4.6.7.1 Mandrel. The mandrel shall be constructed of polytetrafluoroethylene (PTFE)

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coated steel rod. The tie-down strap shall be tested on mandrel sizes that are 1.000 inch outside diameter and 0.500 inch outside diameter.

4.6.8 Flammability test (see 3.7.7). A single wrapped tie-down strap shall be installed around the center of the mandrel. The mandrel assembly shall be clamped in the specimen holder of the test chamber at an angle of 60 degrees from horizontal angled away from the rear wall of the enclosure with the fastener head oriented towards the opening of the enclosure. The burner shall be adjusted to deliver an all-blue flame, 2 inches high, at a temperature of 1751 °F ±54 °F (955 °C ±30 °C), as measured with a thermocouple pyrometer. With the burner held at a 90 degree angle to the longitudinal axis of the mandrel and the long dimension of the flame spreader parallel to the longitudinal axis of the mandrel, the hot tip of the flame shall be applied to the tie-down strap so that the midpoint of the flame touches the fastener head on the assembled specimen. The period of flame application shall be 30 seconds. At the close of the application period, the flame shall be withdrawn and the duration of the after flame shall be recorded.

4.6.8.1 Apparatus. The flammability test chamber shall be approximately 1 foot square by 24 inches in height, and shall be open at top and front to provide required ventilation for combustion but prevent drafts. A mandrel of a .750 inch nominal diameter constructed of non-flammable metallic material shall be used to mount the test specimen. Means shall be provided in the chamber to hold the mandrel at an angle of 60 degrees from horizontal in a vertical plane parallel to and about 6 inches in front of the rear wall of the chamber. The test burner shall be an ASTM D5025 Bunsen type gas burner having a .250 inch inlet, a needle valve in the base for gas adjustment, a bore of .375 inch nominal, and a length of approximately 4 inches above the air inlets. The burner shall be fitted with a wing top flame spreader having a .0625 inch by 2 inch opening.

4.6.9 Product legibility (see 3.7.8). Product identification markings shall be visually examined for legibility. Ten X magnification assistance to the human eye is allowed.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The tie-down straps are intended for use in general wire and cable bundle retention. The tie-down straps are designed to be used in high vibration, shock, highly corrosive, and high temperature environments unique to military applications.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Tie-down strap part number (see 1.2).
c. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 32554 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the Commander, Naval Air Systems Command (Code 4.4.5.3), 48298 Shaw Road, Building 1461, Patuxent River, MD 20670-1900 or QPL_Support@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Qualification inspection process. At the initiation of the qualification process, prior to testing, prospective suppliers must forward a written request for qualification to the qualifying activity (see 6.3). The qualifying activity will respond in writing with a formal "letter of authorization" providing detailed instructions for the submission of product samples. Testing cannot begin until the supplier has received this authorization letter. The qualification process is further described in the SD-6 "Provisions Governing Qualification." Copies of this document are available online at https://assist.dla.mil.

6.3.2 QPL supplemental information. The qualifying activity provides a summarized list of all qualified sources on a publicly accessible website http://www.navair.navy.mil/qpl/. The summary includes but is not limited to the supplier approved part number and related specification part number, a dedicated approval reference number, a supplier location where purchases may be requested and the manufacturing location of the component.

6.4 Installation of MIL-DTL-32554 tie-down straps. Use of the MIL-DTL-32555 tool is recommended to ensure that the tie-down strap is correctly tensioned and not over tightened around the wire bundle. The fastener is designed to lock at a set minimum tension. Additional tensioning beyond the minimum does not improve the holding capability of the tie-down strap and may damage the wire bundle. The tie-down strap may be installed using multiple wraps of the strap around the wire bundle to achieve increased tensile strength as required (see table IV). Additional wraps may provide additional support for larger bundles and reduce compression of coaxial cables by increasing the surface area of the strap around the cable/wire harness. The process of applying traditional lacing tape results in two wraps of the lacing tape around the cable/wire bundle to form the clove hitch knot to complete the installation.

6.5 Subject term (key word) listing.

Clamping cables
Fastening aircraft cables
Fastening device
Wire harness

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TABLE I. Tie-down strap material.

Lace Material Designator	Lace Material	Strap Color Designator	Strap Color
-1	A-A-52084-2-C	-1	White
-1	A-A-52084-2-C	-2	Black

TABLE II. Tie-down strap dimensions.

Length Designator	Strap length L (in.) +0.35, -0.00	Fastener Length A (in.) Max	Fastener Width B (in.) Max	Fastener Height C (in.) Max	Weight (oz.) Max
06	6.00	0.300	0.250	0.200	0.011
10	10.00	0.300	0.250	0.200	0.013
18	18.00	0.300	0.250	0.200	0.018
24	24.00	0.300	0.250	0.200	0.021
44	24.00	0.300	0.250	0.200	0.030

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TABLE III. Qualification inspection.

Examination or Test	Requirement Paragraph	Test Paragraph (in sequence)	Number of Samples
Visual examination 1/	3.6, 3.7.8, 3.7.9, 3.7.10	4.5.1	10
Tensile strength 1/	3.7.1	4.6.1	10
Temperature cycling	3.7.2	4.6.2	10
Tensile strength after temperature cycling 1/	See table V	4.6.3	10
Vibration	3.7.3	4.6.4	
Fluid immersion	3.7.4	4.6.5	10
Fuel immersion	3.7.5	4.6.6	10
Slip resistance 1/	3.7.6	4.6.7	10
Flammability	3.7.7	4.6.8	10
Product legibility 1/	3.7.8	4.6.9	4

1/ Retention of qualification inspections to be conducted by the qualifying activity.

TABLE IV. Minimum tensile strength.

No. of Wraps Around Mandrel	Tensile strength (lbf)
1	39
2	98
3	164

TABLE V. Minimum tensile strength after temperature cycling.

No. of Wraps Around Mandrel	Tensile strength (lbf)
1	36
2	85
3	164

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TABLE VI. Fluid table.

Test Fluid	Test Temperature (°F)	Immersion time (hours)
MIL-PRF-5606 (Inactive for New Design), Hydraulic Fluid, Petroleum Base, Aircraft Missile, and Ordnance	118-122	4
MIL-PRF-7808, Lubricating Oil, Aircraft Turbine Engine, Synthetic Base		
TT-I-735, Isopropyl Alcohol		

TABLE VII. Minimum tensile strength after fluid immersion.

No. of Wraps Around Mandrel	Hydraulic Fluid (lbf)	Lubricating Oil (lbf)	Isopropyl Alcohol (lbf)	Fuel (lbf)
1	38	41	42	-
2	94	96	103	200
3	152	149	163	-

TABLE VIII. Minimum slip resistance.

No. of Wraps Around Mandrel	1.000 inch mandrel (lbf)	0.500 inch Mandrel (lbf)
1	1.30	2.23
2	2.55	3.18
3	2.24	3.43

TABLE IX. Conformance inspection.

Examination and Tests	Requirement Paragraph	Test Paragraph	Sampling plan
Visual examination	3.6, 3.7.8, 3.7.9, 3.7.10	4.5.1	4.4.1.2
Tensile Strength	3.7.1	4.6.1	
Temperature Cycling	3.7.2	4.6.2	
Product Legibility	3.7.8	4.6.9	

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TABLE X. Zero defect sampling plan.

Lot size	Minimum number of tie-down straps to be tested 1/
1 – 4	All
5 – 50	5
51 – 90	7
91 – 150	11
151 – 280	13
281 – 500	16
501 – 1200	19
1201 – 3200	23
3201 – 10000	29
10001 – 35000	35

1/ For lot sizes that are less than the required sample size, all tie-down straps shall be tested.

TABLE XI. Random vibration profiles by bundle diameter.

Break points for random vibration profile						
Clamp Size	Bundle Diameter (inches)	Overall Acceleration [g_{RMS}]*	20 Hz Breakpoint [g^2/Hz]	500 Hz Breakpoint [g^2/Hz]	800 Hz Breakpoint [g^2/Hz]	Duration [minutes]
Random vibration vertical						
-16	1	18	0.604	0.604	0.0074	17
-24	1.5	8	0.119	0.119	0.0014	45
-32	2	7	0.091	0.091	0.0011	24
Random vibration lateral						
-16	1	18	0.604	0.604	0.0074	28
-24	1.5	12	0.266	0.266	0.0030	20
-32	2	8	0.119	0.119	0.0014	28

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FIGURE 1. Installed tie-down strap.

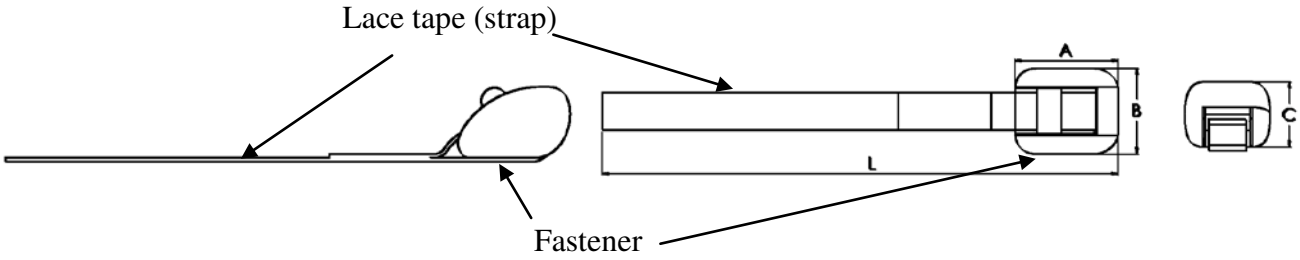


FIGURE 2. Tie-down strap assembly.

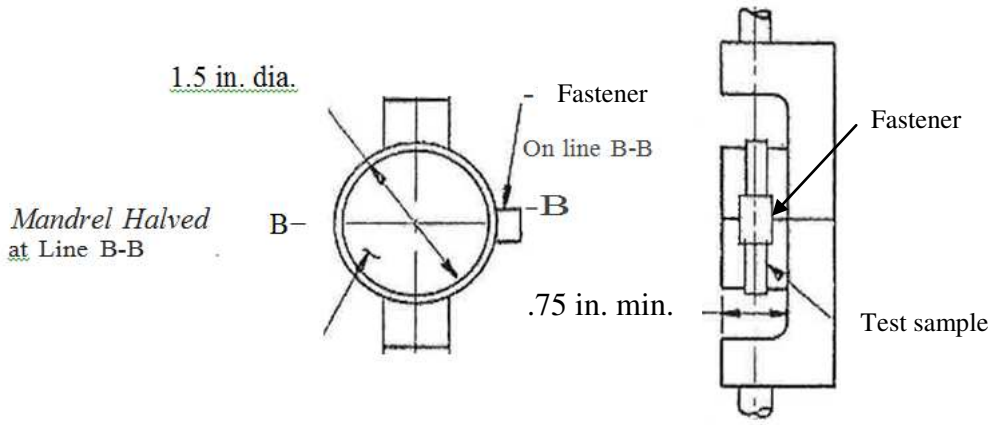


FIGURE 3. Tensile test fixture for strap.

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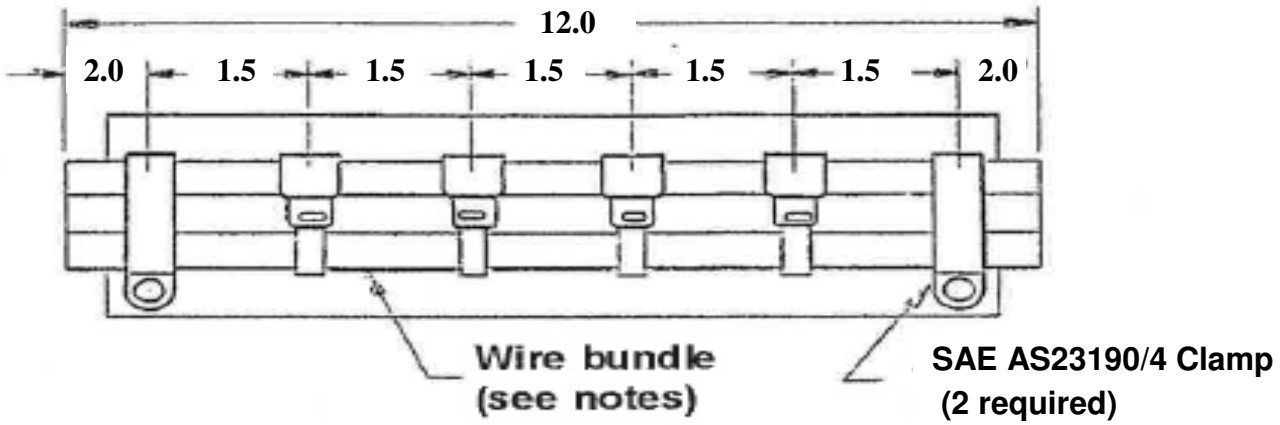


FIGURE 4. Temperature cycling test fixture for tie-down straps (top view).

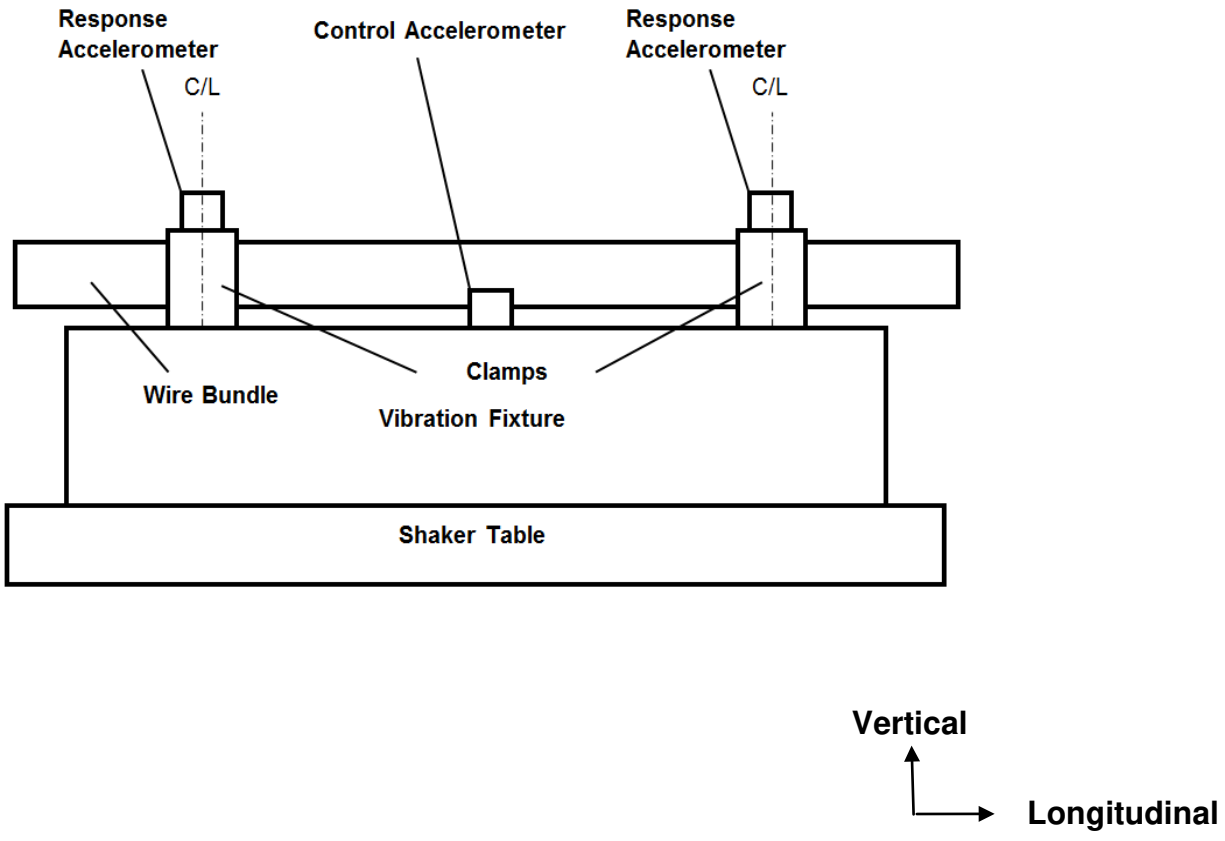
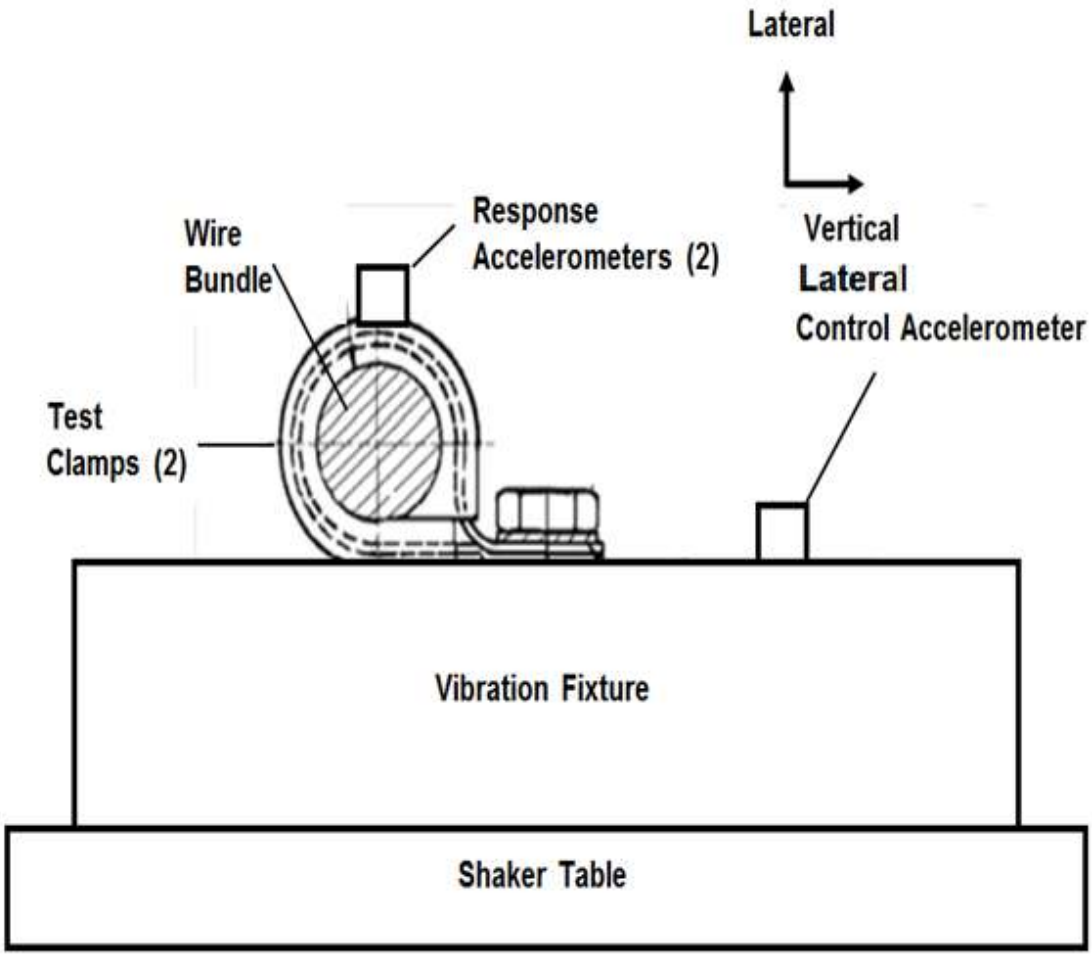


FIGURE 5. Vibration test configuration (side view).

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FIGUR E 6. Vibration test configuration (end view).

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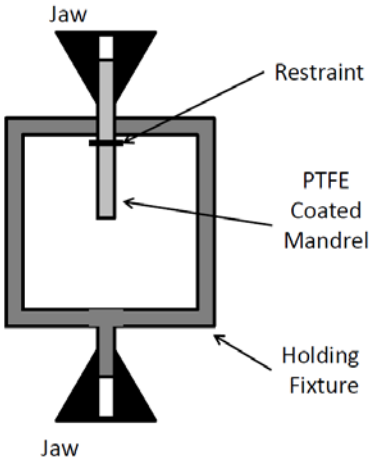


FIGURE 7. Slip resistance test configuration.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85

Preparing activity:
Navy-AS

Project 5975-2016-005

Review activities:
Army - AV, MI
Navy - EC, OS, SH
Air Force - 11, 20
DLA - GS

Industry association:
SAE AE-8C2

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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